Custom Robot Arm-Based Trowel Direction Determination on Wall Corners using Deep Predictive Learning

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***Abstract*—The robot arm is considered because the wrist joint is particularly effective for plastering tasks in construction. Moreover, the present study simulates expert plasterers' actions to design a deep predictive learning model that captures the difference between the current action and the predicted action, including visual information. Deep predictive learning is implemented using a convolutional neural network (CNN) to extract features, with some positional data inserted into the CNN model after feature extraction. This study does not focus on controlling the robot arm itself, but rather on determining whether the trowel direction should change as the robot arm approaches a wall corner. The effectiveness of this approach was validated: as the custom robot arm nears a wall corner, the trowel direction adjusts accordingly.**

***Index Terms*—Custom robot arm, Deep predictive learning, Trowel direction on wall corner**

I. Introduction

Robots have been applied in various fields, including plastering robots in construction [1], and warehousing and logistics [2]. In the construction fields, joints such as the knee, wrist, and arm must be considered. However, the wrist joint is particularly effective for plastering tasks [3]. Therefore,in this study, only the robot arm is considered. Several studies have proposed plastering systems using robotic arms [4], [5]. Some plastering recognition systems focus solely on determining whether the plastering task has been completed, without considering the painting tools or materials used [6], [7]. Du et al. [8] proposed a plastering location recognition system that takes both the painting tool and materials into account. Their recognition system not only determines whether the plastering task should continue or stop, but also explicitly considers the specific painting tools and materials involved.

Moreover, the robot arm plasters the wall based on instructions provided by these recognition system. Although reinforcement learning has been applied to path planning or assembly planning [9]. Such reinforcement learning approaches require controlled design environments and involves tens of thousands of training trials. Therefore, reinforcement learning is currently difficult to apply to plastering work. Additionally, an initial-stage deep learning model was proposed in [10]. Deep predictive learning is defined and described in [11], [12]. It simulates human behavior by measuring difference between predictions and actual outcomes. Based on this, this study simulates expert plasters' actions to design a model that captures the difference between the current action and the predicted action, including vision information. Using the predictive learning model, the robot arm's movements can be optimized under different conditions.

Expert plasterers typically employ three main techniques when painting walls: vertical, horizontal, and smoothing [13]. Initially, they prioritize covering large areas with vertical strokes, then use horizontal and smoothing techniques to evenly coat the entire wall with plaster. In contrast, a robot arm equipped with a recognition system typically repeats vertical strokes until the wall is fully coated. Although achieving coverage, the robot's final result differs from that of skilled plasterers. Therefore, incorporating horizontal and smoothing techniques becomes crucial. In particular, the direction of the trowel must vary depending on the area of the wall, especially when approaching corners. Thus, this study focuses on determining the optimal trowel direction at wall corners.

Importantly, this study does not focus on controlling the robot arm itself, but rather on determining whether the trowel direction should change as the robot arm approaches a wall corner. To address this, a custom robot arm was developed based on [14], incorporating a new capability: trowel rotation. Deep predictive learning is implemented using a convolutional

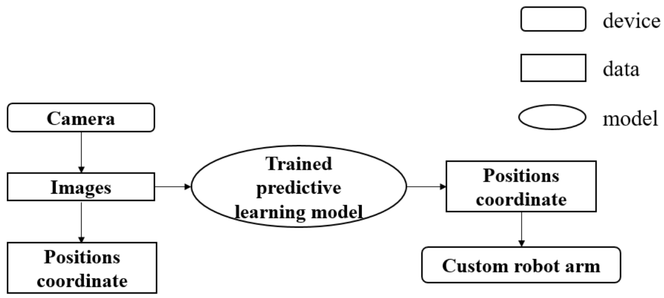

Fig. 1. Flowchart of the proposed method

neural network (CNN) to extract features, with some positional data inserted into the CNN model after feature extraction. Training data were collected from videos of expert plasterers to determine the optimal trowel directions. The effectiveness of this approach was validated: as the custom robot arm nears a wall corner, the trowel direction adjusts accordingly. The main contributions of this paper are as follows:

1) Designing an effective deep predictive model to predict robot arm movements under different conditions using expert plasterer videos.
2) Developing a custom robot arm with the capability to rotate the trowel.

This paper is structured as follows: after the introduction (Section I), the proposal is explained (Section II). The proposal will be validated through experiments (Section III), followed by the conclusion (Section IV).

II. Our Proposed Method

The proposal is illustrated in Fig. 1. Images capturing the conditions of painted walls are obtained from a camera. These images are then input into a trained predictive learning model to predict position coordinates. Subsequently, the custom robot arm adjusts the trowel rotation based on these coordinates when encountering wall corners. The custom robot arm is described in Section II.A, while the structure of predictive learning is explained in Section II.B.

A. A custom robot arm design

A custom 4-axis robot arm designed for simulating plastering work, based on the ABB IRB 460 model [15] and Dong's custom plastering robot arm [14], is shown in Fig. 2. This study incorporates a rotatable trowel design. A motor is inserted at the end of the arm to operate the trowel, creating a simple rotatable trowel mechanism.

B. The structure of a deep predictive learning

The proposed model imitates plaster actions and visuals to design the training data and model. Visual data for plastering is extracted from the plastering video shown in Fig. 3, while action data is derived from the trowel depicted in Fig. 4. Fig. 3 shows the rotation of the trowel as it approaches the wall corners and showcases images extracted at 0.008333-second intervals from a 5-minute video, totaling 36,001 images, each

Fig. 2. The custom robot arm with a trowel

sized 160 × 120 pixels. The plastering action is represented by a 12-dimensional vector, comprising 3 dimensions for the origin coordinates on the wall (X, Y, Z) and 9 dimensions for the position of the trowel's left corner (xx, xy, xz, yx, yy, yz, zx, zy, zz), which indicate the orientation of the trowel.

The proposed model integrates trowel rotation when facing a wall corner, as illustrated in Fig. 5. Initially, each video frame image serves as input data fed into a Convolutional Neural Network (CNN) [16] to extract visual features, denoted as $F(i)_{visual}$ in Equation (1):

$$F(i)_{visual} = CNN(image(i)) \tag{1}$$

The CNN architecture consists of: Three convolutional layers, each followed by Batch Normalization [17], ReLU activation [18], and Max Pooling with a stride of 2:

1) First layer: 16 filters of size 3×3.
2) Second layer: 32 filters of size 3×3.
3) Third layer: 64 filters of size 3×3, followed by Adaptive Average Pooling to achieve a fixed output size of 8×6.

After flattening, the resulting feature vector has a dimension of 3072.

Next, the 12-dimensional position vector of the plaster action undergoes transformation via a Fully Connected layer (FC), including linear transformation, Batch Normalization, and ReLU activation, yielding $FC(i)_{vector}$ as the current position vector in Equation (2):

$$F(i)_{spatial} = FC(i)_{vector} \tag{2}$$

These visual and spatial feature vectors are concatenated into $F(i)$ as a unified feature vector of dimension 3328:

$$F(i) = Concat(F(i)_{visual}, F(i)_{spatial}) \tag{3}$$

This combined feature representation is then processed through a series of fully connected layers:

1) The first FC layer reduces dimensionality to 128 with ReLU activation.
2) The final linear layer outputs a 12-dimensional vector, predicting the next position (i+1) of the trowel to guide robotic arm movement.

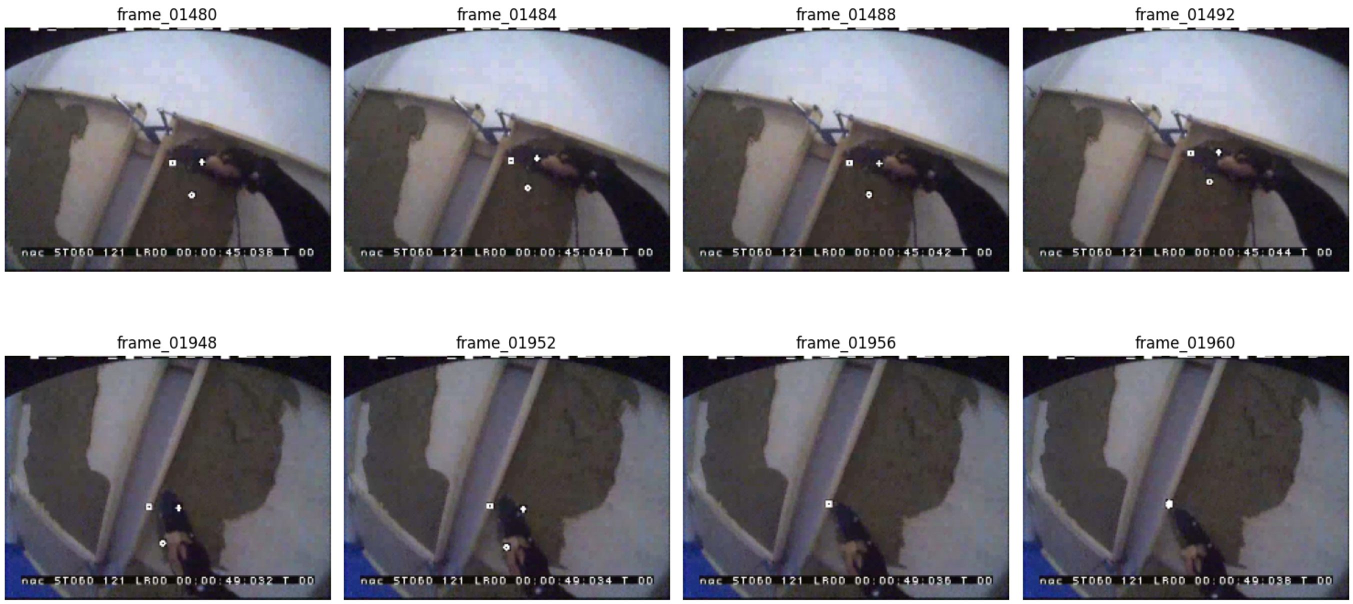

Fig. 3. Video data for the train model

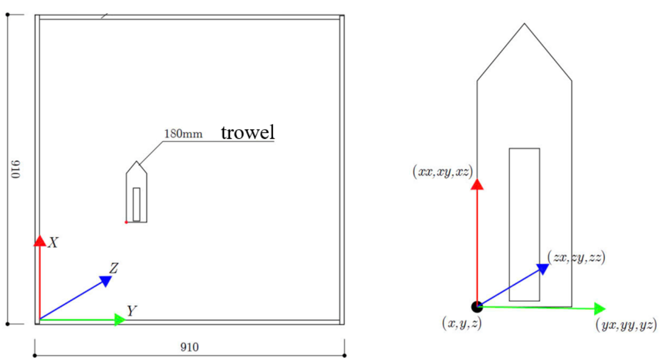

Fig. 4. Action data for the train model

III. Experiment

The proposed model was trained on a computer equipped with an AMD Ryzen 7 5800H (16 CPU cores) running at 3.2 GHz, an NVIDIA GeForce RTX 3050 GPU, and 16GB of RAM, ensuring adequate computational performance. It utilizes video data (Section III.A) and incorporates custom robot arm-based trowel direction changes on wall corners (Section III.B) to validate the effectiveness of the proposed model.

A. Video data from a plaster

The proposal involves two processes: train and test using video data of plaster. A total of 36,001 images are extracted from the video data of the plaster. In the train process, 80% of the data is used to train the model, while the remaining 20% is reserved for test.

The model was trained using the following parameters: 50 epochs, a batch size of 32, a learning rate of 0.001, a weight decay of 0.0004, AdamW optimizer, and an $L1$ loss function. Fig. 6 shows the train and test losses of the model with 50 epochs. Both processes converged as depicted in Fig. 6.

The proposed model is evaluated using test data from videos of plaster, which capture the position and rotation of the trowel with coordinates expressed in 12 dimensions. Although the origin coordinates on the wall (X, Y, Z) are changed relative to the position of the trowel, (X, Y, Z) is defined as the position of the trowel. The position of the trowel's left corner (xx, xy, xz, yx, yy, yz, zx, zy, zz) is defined as the rotation of the trowel. Fig. 7 shows the position of the trowel, while Fig. 8 illustrates its rotation. The following are three ways to evaluate the trained model:

1) **Actual vs. Predicted Data for Trowel Position and Rotation in One Dimension:**
20% of the data were used as test data to predict the position and rotation of the trowel. The original 12-dimensional data consist of the first 3 dimensions representing the trowel's position, while the remaining 9 dimensions correspond to its rotation. These 3-dimensional position data and 9-dimensional rotation data were each transformed into one-dimensional data separately. Fig. 7(a) and 8(a) display the actual and predicted values of the trowel's position. When these values are close, they align along the red line. The range

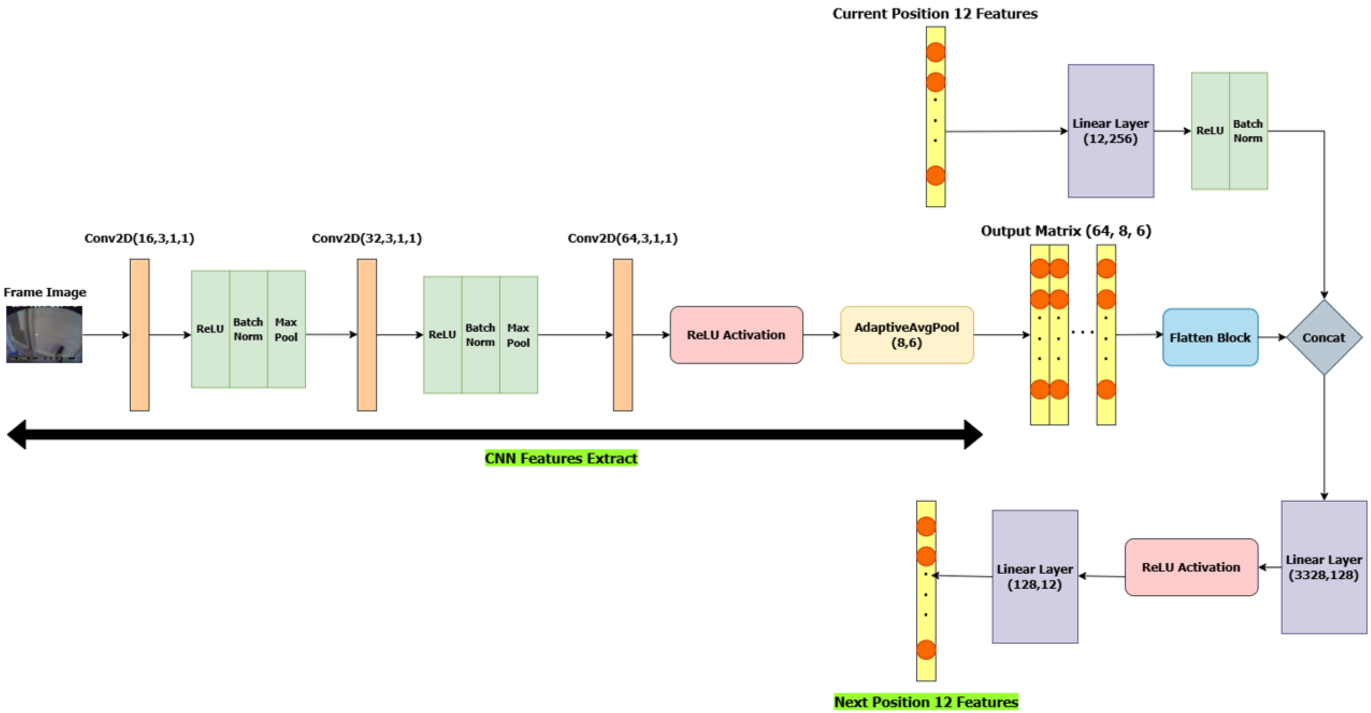

Fig. 5. The train model structure

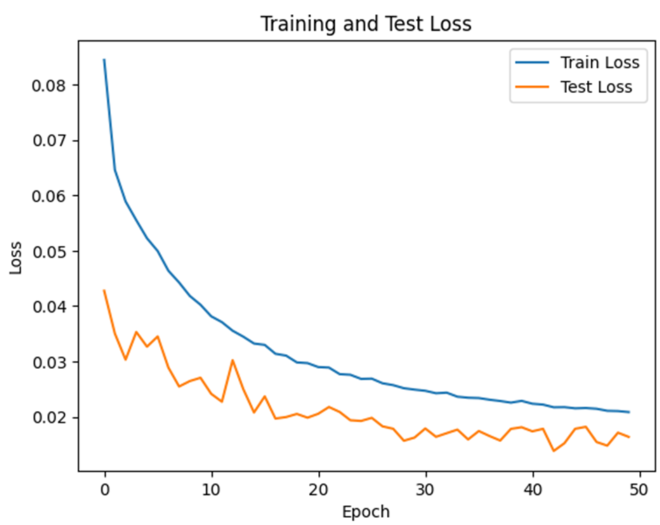

Fig. 6. Loss function for the train and test data

of Fig. 7(a) spans from -1300 to 1300, while the range of Fig. 8(a) spans from -1 to 1. From both figures, it can be observed that the predicted values closely match the actual values, indicating that the test data were predicted with high accuracy.

2) **Actual vs. Predicted Data for Trowel Position and Rotation:**
Fig. 7(b) shows the relationship between the data index and the first three dimensions of the trowel's position, constrained within the wall size. The predicted and actual values are closely aligned. The data points in the 3D coordinate space are tightly clustered, indicating minimal spatial movement. For rotation, the trowel's orientation is represented using a 9-dimensional vector corresponding to a rotation matrix. This rotation matrix can be computed using Rodrigues' rotation formula. Fig. 8(b) illustrates the data index against various rotation angles. Again, the predicted and actual values closely match, indicating that the trained model can effectively predict the trowel's rotation.

3) **Differences Between Actual and Predicted Data for Trowel Position and Rotation:**
The differences between actual and predicted trowel positions are shown in Fig. 7(c). These differences are calculated as the root of the sum of squared differences between each coordinate of the actual and predicted values. For all test data, the positional differences are less than 50 mm, indicating that the predicted values are closely aligned with the actual values. Fig. 8(c) shows the differences in rotation between actual and predicted trowel orientations. These differences are computed based on the angular difference between the actual and predicted rotations. For all test data, the angular differences are close to zero, suggesting that the model accurately predicts the trowel's rotation.

B. Custom robot arm-based trowel direction determination

The trained model is applied to a custom robot that can change the trowel direction when the trowel faces a wall

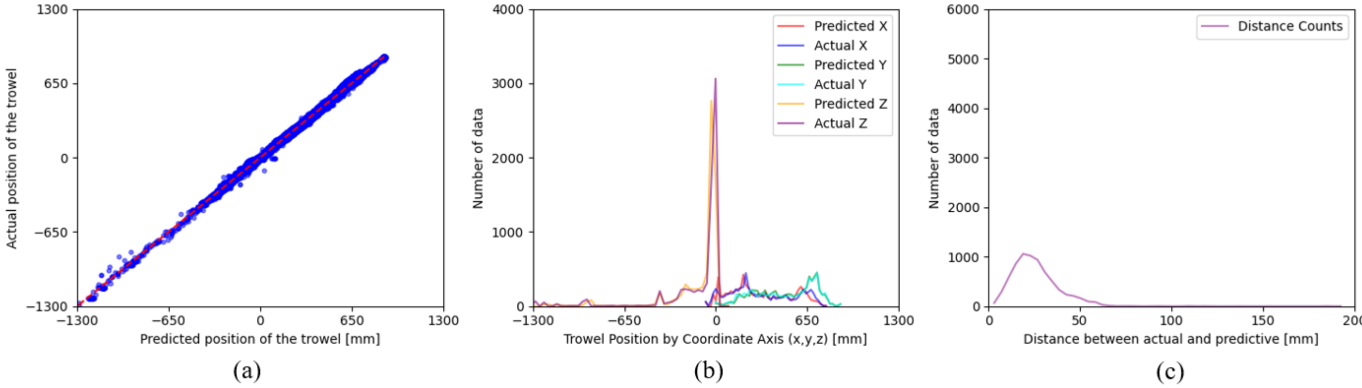

Fig. 7. The position of the trowel

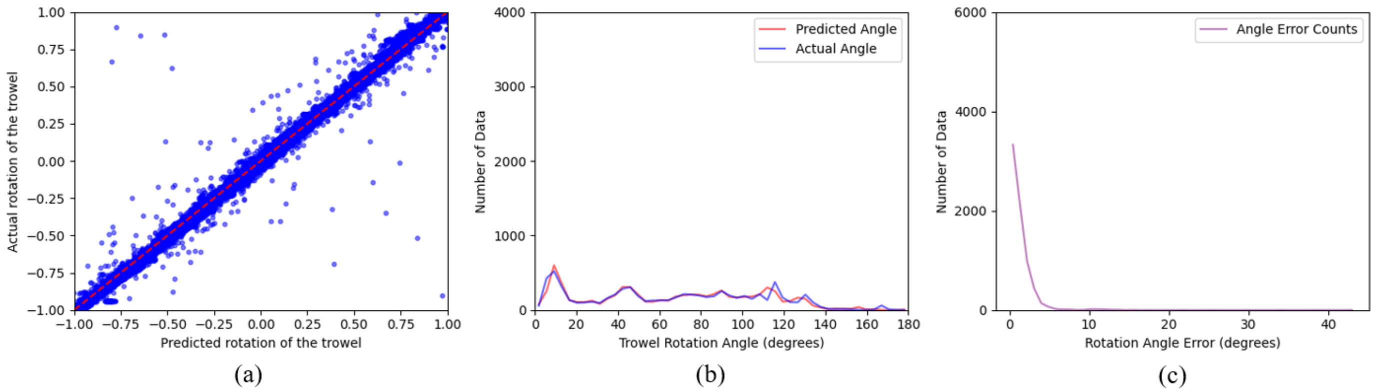

Fig. 8. The rotation of the trowel

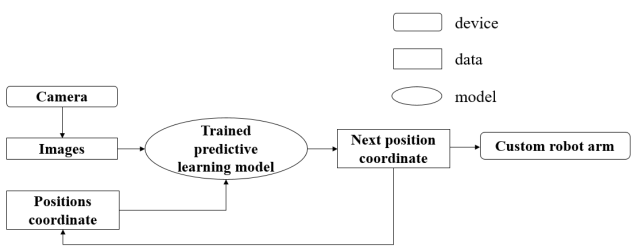

Fig. 9. Flowchart of custom robot arm

corner, as shown in Fig. 9. Fig. 9 is similar to Fig. 1, with the following differences:

1) **Input data:** Images are captured in real-time using an OAK-1 camera. Position coordinates are measured by imitating the training data.
2) **Output data:** The next position coordinates serve two purposes—one is sent to the custom robot arm, and the other is used as input for the trained model.

IV. Conclusion

This study presents the design of a custom robotic arm equipped with a rotational trowel that mimics the motion of a plastering wrist. In addition, a deep predictive learning model based on Convolutional Neural Networks (CNNs) and positional data is proposed. The effectiveness of the trained model is evaluated using plastering data and experiments with the custom robotic arm and trowel. In future work, the trained model will be applied to the robot arm to predict how to perform plastering on walls under varying conditions, such as around corners or edges, using the trowel.

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References

[1] L. Rampini, A. Khodabakhshian, F. Cecconi, “Artificial Intelligence Feasibility in Construction Industry,” EC3 Conference 2022. Vol. 3. European Conference on Computing in Construction 2022.

[2] J. Zhao, B. Luo, H. Meng, Q. Li, “Research on Intelligent Warehousing Based on AGV Handling Robot,” 2023 5th International Conference on Frontiers Technology of Information and Computer (ICFTIC), Nov. 17-19, 2023, Qiangdao, China.

[3] I. Noborio, M. Muramoto, T. Nishimura, “Motion Analysis of Traditional Earth Plastering for the Development of Plastering Robots,” Research Report of the Kinki Chapter, Architectural Institute of Japan. Structural Section, Vol. 61, pp. 1-4, June 2021. (in Japanese)

[4] Z. Liu, D. Chen, M. Eldosoky, Z. Ye, X. Jiang, Y. Liu, S. Ge, “Puttybot: A sensorized robot for autonomous putty plastering,” Journal of field robotics, 2024.

[5] Y. Wang, L. Xie, J. Chen, M. Chen, T. Hu, H. Liao, S. Sun, J. Chen, “Mortar Spraying and Plastering Integrated Robot for Wall Construction,” Automation in Construction, Vol. 165, Sep. 2024, 10553.

[6] M. Amari, S. Abdullah, L. Sheng, L. Aik, “Development of 4 Axis Shadow Robot Arm Integrated With Flex Sensor Mechanism,” International Journal of Mechanical Engineering and Robotics Research, Vol. 9, No. 3, pp. 441-446, March 2020.

[7] J. Kongthon, “Tracking Improvements for Nonlinear Systems Using Inversion-Based Feedforward Control and Feedback Control: A Robotic Arm Example,” International Journal of Mechanical Engineering and Robotics Research, Vol. 9, No. 2, pp. 225-232, February 2020.

[8] W. Du, R. Morimoto, I. Noborio, M. Muramoto, “Plastering Location Recognition System with a Robot Arm,” 11th International Conference on Computing and Artificial Intelligence March pp. 28-31, 2025, Kyoto, Japan.

[9] A. Zhu, T. Dai et al., “Deep Reinforcement Learning for Real-Time Assembly Planning in Robot-Based Prefabricated Construction,” IEEE Transactions on Automation Science and Engineering. Vol. 20, No. 3, July, 2023, pp. 1515-1526.

[10] K. Noda, H. Arie, Y. Suga, T. Ogata, “Multimodal Integration Learning of Robot Behavior using Deep Neural Networks,” Robotics and Autonomous Systems, Vol. 62, No. 6, pp. 721-736, 2014.

[11] R. O’Reilly, D. Wyatte, J. Rohrlich, “Deep Predictive Learning: A Comprehensive Model of Three Visual Streams,” arXiv preprint arXiv:1709.04654, 2017.

[12] R. O’Reilly, J. Russin, M. Zolfaghar, J. Rohrlich, “Deep Predictive Learning in Neocortex and Pulvinar,” Journal of Cognitive Neuroscience, Vol. 33, No. 6, pp. 1158-1196.

[13] I. Noborio, M. Muramoto et al., “Development of a Soil-Plastering Robot Using Motion Capture Data of Plasterers,” Proceedings of the 44th Symposium on Information, Systems, Utilization, and Technology, Architectural Institute of Japan, pp. 599-602, vol. 44, 2021. (in Japanese)

[14] S. Dong, M. Muramoto, W. Du, ”A Custom Robotic Arm Simulation Using RoboDK for Plastering Work,”Proceedings of the 2024 10th International Conference on Computing and Artificial Intelligence, pp. 228-232.

[15] ABB Robotics. n.d. IRB 460 Articulated Robots. Retrieved from https://new.abb.com/products/robotics/robots/articulated-robots/irb-460.

[16] K. O’Shea, R. Nash, ”An introduction to convolutional neural networks,” 2015. [Online]. Available: https://arxiv.org/abs/1511.08458. [Accessed: Mar. 20, 2025].

[17] S. Ioffe, C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” in Proceedings of the 32nd International Conference on Machine Learning (ICML), 2015, pp. 448–456.

[18] A. F. M. Agarap, ”Deep Learning using Rectified Linear Units (ReLU),” arXiv preprint arXiv:1803.08375, 2018.